AGE 310: Instrumentation and Measurements

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Abstract

Lecture 10 focuses on the performance of instruments when the measured quantity is changing rapidly, known as Dynamic Characteristics. We mathematically model instruments as systems of Zero, First, or Second Order, analyzing their response to a sudden change (Step Input). Mastery of Time Constant (τ) and Damping Ratio (ζ) is essential for selecting sensors appropriate for high-speed, dynamic agricultural processes, such as engine monitoring or robotic motion control.

Lecture 10: Dynamic Characteristics of Instruments

10.1 Introduction to Dynamic Response

10.1.1 Static vs. Dynamic

- **Static Response:** Describes the relationship between the sensor's input and output when the input is unchanging (steady state). (Covered in Lecture 1: Accuracy, Sensitivity).
- **Dynamic Response:** Describes the behavior of the instrument when the input changes abruptly or rapidly over time. The output will lag or overshoot the input due to the sensor's physical limitations (inertia, capacitance, thermal mass).

10.1.2 Modeling Instruments

Most instrument systems can be accurately modeled using ordinary linear differential equations, where the order (n) is defined by the highest derivative present.

$$a_n \frac{d^n y}{dt^n} + \cdots + a_1 \frac{dy}{dt} + a_0 y = b_0 x(t)$$

Where $x(t)$ is the input (measurand) and $y(t)$ is the output (reading).

10.2 Zero Order Systems ($n = 0$)

10.2.1 Characteristics

A Zero Order system contains **no energy storage elements** (no mass, no thermal capacity, no capacitance).

$$a_0 y = b_0 x(t) \quad \Longrightarrow \quad y(t) = K \cdot x(t)$$

Where $K = b_0/a_0$ is the Static Sensitivity.

- **Response:** The output is **instantaneous** and proportional to the input. There is no lag.
- **Examples:** Ideal Potentiometer (displacement, Lecture 3), Resistor (Ohm's law).

This idealization holds true only for low-frequency inputs where inertia is negligible.

Calculation Example 1: Dynamic Error (Zero Order)

A pressure sensor is modeled as a Zero Order system with a sensitivity $K = 0.5$ V/kPa. If the pressure instantaneously jumps from 100 kPa to 120 kPa (a Step Input), calculate the Dynamic Error after 1 µs.

Solution: The output is instant and exactly tracks the input.

- $\text{Output}_{\text{ideal}} = 0.5$ V/kPa $\times\ 120$ kPa $= 60$ V.
- Since the response is instantaneous, the reading at 1 µs is 60 V.
- Dynamic Error $=$ True Output $-$ Measured Output $= 60$ V $- 60$ V $= 0$.

10.3 First Order Systems ($n = 1$)

10.3.1 Characteristics

A First Order system contains **one energy storage element** (e.g., thermal capacity of a thermometer, resistance-capacitance (RC) circuit). The output cannot change instantly, leading to exponential lag.

$$\tau \frac{dy}{dt} + y = Kx(t)$$

Where τ (tau) is the **Time Constant**.

10.3.2 The Time Constant (τ)

The Time Constant (τ) is the measure of the system's sluggishness or speed of response. It is defined as the time required for the system output to reach **63.2%** of its final steady-state value after a step change.

Step Response Equation (to a value Y_f):

$$y(t) = Y_i + (Y_f - Y_i)(1 - e^{-t/\tau})$$

- After 1τ, the system is 63.2% complete.
- After 3τ, the system is 95.0% complete.
- After 5τ, the system is 99.3% complete (often considered to be at steady state).

Examples: Most thermometers (RTD in a protective sheath), RC electrical circuits, simple pressure gauges.

Calculation Example 2: First Order Response (Thermometer)

A grain silo temperature sensor (RTD) is modeled as a First Order system with a time constant $\tau = 60$ seconds. It is initially reading $25°$C. The actual temperature abruptly jumps to $50°$C (Step Input).

1. Calculate the temperature reading after 60 seconds (1τ).
2. Calculate the time required to reach 95% of the final change.

Solution:

1. **Reading after 60 s:** After 1τ, the output reaches 63.2% of the total change ($50°\text{C} - 25°\text{C} = 25°\text{C}$).
$$\Delta T = 0.632 \times 25°\text{C} \approx 15.8°\text{C}$$
$$T_{\text{reading}} = T_{\text{initial}} + \Delta T = 25°\text{C} + 15.8°\text{C} = 40.8°\text{C}$$
2. **Time to 95% completion:** This occurs at 3τ.
$$t_{95\%} = 3 \times \tau = 3 \times 60 \text{ s} = 180 \text{ seconds or } 3 \text{ minutes}$$

10.4 Second Order Systems ($n = 2$)

10.4.1 Characteristics

A Second Order system contains **two energy storage elements** (e.g., mass, spring, and damper (friction); or inductance (L) and capacitance (C)). The system can oscillate before settling.

$$\frac{1}{\omega_n^2}\frac{d^2y}{dt^2} + \frac{2\zeta}{\omega_n}\frac{dy}{dt} + y = Kx(t)$$

Where:

- ω_n: **Natural Frequency** (rad/s) - How fast the system wants to oscillate.
- ζ (zeta): **Damping Ratio** - Measures how quickly the oscillation fades.

[Image of Diagram of a mass-spring-damper system illustrating inertia and damping]

10.4.2 Damping Ratio (ζ)

The value of ζ determines the nature of the dynamic response:

1. **Underdamped ($\zeta < 1$):** The output overshoots the final value and oscillates exponentially before settling. Provides the fastest response but introduces dynamic error (overshoot).
2. **Critically Damped ($\zeta = 1$):** The system reaches the final value in the minimum time possible without any oscillation or overshoot. This is the ideal response for most instrumentation.
3. **Overdamped ($\zeta > 1$):** The system is sluggish and approaches the final value slowly without oscillation.

Examples: Accelerometers, Pressure transducers with heavy diaphragms, mechanical gauges, LVDTs.

Calculation Example 3: Dynamic Overshoot

An accelerometer used in a robotic weeding tool is modeled as a Second Order system. It has a natural frequency $\omega_n = 500$ rad/s and a damping ratio $\zeta = 0.5$ (Underdamped).

The Percentage Overshoot (%OS) is calculated by:

$$\%OS = 100 \cdot \exp\left(\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}\right)$$

Calculate the Percentage Overshoot when a sudden force is applied.

Solution:

$$\%OS = 100 \cdot \exp\left(\frac{-\pi \cdot 0.5}{\sqrt{1-0.5^2}}\right)$$

$$\%OS = 100 \cdot \exp\left(\frac{-1.5708}{\sqrt{0.75}}\right) = 100 \cdot \exp(-1.8137) \approx 16.3\%$$

This means the measured acceleration will temporarily read 16.3% higher than the true steady-state acceleration before oscillating down.

10.5 Trade-offs and Selection

10.5.1 Dynamic Error vs. Measurement Lag

- **Overshoot (Underdamped):** Causes transient dynamic error (the reading is temporarily too high or too low).
- **Lag (Overdamped or First Order):** Causes measurement lag (the reading is behind the true value).

In instrumentation, especially in control systems, the preference is for the critically damped ($\zeta = 1$) response, as it provides the fastest possible signal without dangerous oscillation.

10.5.2 Selection Criteria

- **Slow Variables (e.g., Soil Temperature):** A sensor with a large τ (long time constant, like a thermometer in a sheath) is acceptable, as the input rarely changes fast.
- **Fast Variables (e.g., Engine Speed, Vibration):** A sensor must have a fast response (small τ) and ideally be critically damped. Using a sluggish sensor will introduce lag that makes real-time control impossible.

10.6 Summary of Key Terms (Swahili/Context)

- **Dynamic Response:** Majibu ya Kubadilika (Response to changing input).
- **Step Input:** Mabadiliko ya Ghafla (Sudden, instantaneous change in input).
- **Order of System:** Ngazi ya Mfumo (Defined by the highest derivative in the model).
- **Time Constant (τ):** Muda Usiokoma (Time for 63.2% completion in First Order systems).
- **Natural Frequency (ω_n):** Masafa Asilia (Rate the system wants to oscillate).
- **Damping Ratio (ζ):** Uwiano wa Kuzima (Measures oscillation decay).
- **Critically Damped:** Kizingiti cha Kuzima (Fastest response without oscillation, $\zeta = 1$).
- **Overshoot:** Kupita Kiasi (Transient error in underdamped systems).

Lecture 10: Review Questions

SECTION A: Basic Concepts (5 Questions)

1. Define the term *Dynamic Characteristic*. Explain why this property is insignificant when measuring the average annual rainfall but crucial when measuring instantaneous drawbar pull of a tractor.
2. What is the defining characteristic of a *Zero Order* instrument? Give an example of a Zero Order sensor studied in previous lectures (idealized).
3. Define the *Time Constant* (τ) of a First Order system. What percentage of the total steady-state change is achieved after 3τ?
4. State the two key physical elements necessary for a measurement system to be classified as a *Second Order* system.
5. Differentiate between *Underdamped* ($\zeta < 1$) and *Overdamped* ($\zeta > 1$) responses in a Second Order system, focusing on stability and speed.

SECTION B: Advanced Analysis and Calculations (10 Questions)

6. **First Order Response Calculation:** A sensor monitoring biogas pressure is rated as a First Order system with $\tau = 10$ seconds. If the pressure suddenly increases from 5 kPa to 15 kPa (a step change), calculate the pressure reading after 25 seconds.
7. **Selection for Control:** An engineer needs to select a temperature sensor for a high-speed chicken incubator control system, where temperature must settle within 1°C of the setpoint (35°C) within 30 seconds. If Sensor A has $\tau = 5$ s and Sensor B has $\tau = 15$ s, which sensor is unsuitable, and why?
8. **Dynamic Error Calculation (First Order):** A flow meter ($\tau = 2$ s) suddenly receives a flow rate of 10 L/s. Calculate the *Dynamic Error* (Difference between True Value and Measured Value) 1 second after the flow begins.
9. **Second Order Damping Selection:** Explain why the *Critically Damped* response ($\zeta = 1$) is the most desired behavior for high-precision mechanical instrumentation like LVDTs and accelerometers. What is the fundamental trade-off made if the system is designed to be $\zeta = 0.5$?
10. **Percentage Overshoot Calculation:** A new tractor suspension system, measured using an accelerometer, is found to have a damping ratio $\zeta = 0.4$. Calculate the percentage overshoot (%OS) of the vertical displacement response when hitting a sudden bump (step input).

$$\%OS = 100 \cdot \exp\left(\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}\right)$$

11. **First Order Time Constant Determination:** An RC circuit used to filter sensor noise has $R = 100$ kΩ and $C = 100$ µF.
    - Calculate the Time Constant (τ) in seconds. ($\tau = RC$).

- Calculate the time required for the circuit output voltage to reach 98% of the input voltage.

12. **Dynamic and Static Error Differentiation:** Differentiate between *Static Error* and *Dynamic Error*. Which one is eliminated by calibration, and which one is eliminated by selecting a faster sensor?

13. **Mass-Spring-Damper Analogy:** In the context of a pressure sensor using a diaphragm, identify the physical components that correspond to the *Mass*, the *Spring*, and the *Damper* elements of a Second Order system.

14. **Order Classification:** Classify the following instruments as Zero, First, or Second Order systems and briefly justify your choice:

    - A mercury thermometer.
    - A DC voltmeter.
    - A complex pressure transducer with a diaphragm.

15. **Lagging Measurement Consequences:** In a closed-loop control system managing pesticide application, the flow sensor has a time constant $\tau = 5$ s. The controller requires flow data every 0.5 s. Explain the severe consequence of the sensor's time constant on the stability and efficiency of the pesticide application rate.